

An IoT Based Smart Water Contamination Monitoring System

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Abstract— In the current era of urbanization, there is a need for effective monitoring and alert the concerned water body maintenance agencies to control the contamination levels in the residential water supply. Because of the increasing population and exponential growth of pollution, this task of quality water supply becomes challenging. At present, the concerned water quality maintenance agencies rely on traditional techniques. In this method, they are collecting samples of suspected water from the tanks and dams. Then send these samples to concerned laboratories for testing the level of contamination in the sample. This is not only a time-consuming process but also a costly affair. Also, with the existing method, dissemination of analyzed real-time data to the concerned stakeholders for taking necessary timely action has become a cumbersome job. To overcome all the issues this novel system is proposed. In this paper, a real-time water contamination monitoring system is developed using Internet of Things (IoT) and Embedded Systems. This system can able to estimate the water quality in residential homes as well water body tanks. The system consists of an Arduino Uno, ESP 8266, Analog to Digital Converter, pH sensor, turbidity sensor, temperature sensor, and other electronic components. With this setup, the water contamination can be estimated by measuring the pH value, turbidity of water and temperature of the water. The measured values at the edge level are then pushed to ThingSpeak cloud server via the ESP8266 wifi module. The accumulated data is analyzed in the cloud and from which the results are accessed via an interfacing device. The results obtained can help determine the quality of the water sample to be tested. The system has achieved the intelligence of data processing and networking of information transfer for automated water quality monitoring. It is distinguished by the advantages of efficiency, accuracy, and frugal use of labour and material resources.

Keywords— *Arduino Uno, ESP 8266, Analog, pH sensor, turbidity sensor, temperature sensor, Thing Speak cloud.*

I. INTRODUCTION

Internet of Things is a revolutionary technology that is rapidly applied to different fields with the support of contemporary wireless sensor networks (WSNs) in which the physical objects treated here as “things” are getting connected via the Internet [1]. This disruptive technology will be changed into “The next Industrial Revolution” of this century. As IoT is the combination of different technologies as well it is operated under different heterogeneous components, this technology is considered to be disruptive. Water contamination monitoring is one of the important applications of IoT. The environmental parameters can be monitored by using IoT technologies in which Water contamination is very important one. The Smart Environment means connecting sensors, actuators and computational devices in a network through Internet of Things technologies such that constant monitoring and detailed understanding of variations in natural environment which comprises of non-human made things like air, water, radiation, land and trees etc. Hammi et al. [2]

quoted that, as the population increases, this leads to the increase in the demand for usage of energy resources, domestic water, maintaining sanitation, quality education, cost-effective health services, and other public services like transportation and parking. The best solution for these challenges is “smart cities”. One of the Smart Environment example is Smart Water Contamination Monitoring and Control (SWCMC) system. A detailed understanding of the SWCMC is shown in figure 1. This system is designed with the help of IoT smart sensors, wireless sensor network principles, and cloud computing techniques. The water contamination is sensed with the help of smart aqua sensors placed at different water resources and these sensors are get connected through a wireless sensor network framework [3]. The collected data from the aqua sensors are communicated via IoT connectivity and communication protocols to the cloud. The cloud-based system is embedded with the data analytics and machine learning techniques will analyses the aggregated data in the cloud and based on the contamination level found, the respective departments will take necessary actions. After the quality check is done the respective department may initiate the water contamination control devices via the IoT based control systems [15] and the actuators present in the control system will minimize the causes for the contamination.

II. PROBLEM STATEMENT

A. About water Quality parameters

Generally, water quality parameters are classified into three types, namely physical, chemical, and biological [4,5]. The physical parameters are Turbidity, Temperature, color, electrical conductivity, taste etc. The chemical parameters are pH, Acidity, alkalinity, chloride, fluoride etc. Whereas the general biological parameters are bacteria, algae. In this proposed system we considered two parameters i.e Turbidity and Temperature from Physical and one parameter i.e pH from chemical category.

B. Turbidity

Turbidity is caused majorly due to suspended materials like clay, silt in water. Because of this water looks cloudy. This turbidity influences the water temperature [6]. Because of this reason we consider the temperature of the water as one of the important parameters in this work. Also, the viscosity, odors, solubility, palatability, and chemical reactions are influenced by temperature [7].

C. The pH and Temperature effect:

The pH of a substance is a measure of its acidity or basicity and is an important parameter in determining the quality of water. It is defined as the negative logarithm of the concentration of hydrogen ions in the solution. [5,8]. The pH values range from 0 to 14. The pH values of water from 0 to 6 indicates acidic nature whereas pH value from 8 to 14

indicates base nature. The pH value of 7 is neutral. At temperatures [14, 17] between 25 °c to 30 °c the pH value of water is normally at 7.

D. Proposed System

The existing system uses traditional sampling methods to determine the contamination level of the water reserved. In these methods, water samples are collected manually from different points of the waterbody present in the reservoir or tank. Then by using chemicals, these collected samples are tested in the laboratory so that the purity of water is estimated by conducting tests related to pH, conductivity, water dissolved oxygen, etc. However, this can often result in human error, and these methods of quality checking may not be possible at all times of the day. Also, the accuracy in quality assessment of the entire water may not possible with the manual method. Since the laboratory test results will be obtained after some days or weeks based on the availability of labs and test personnel. Hence, to overcome this issue a real-time water quality measurement automatic system is required, which is used to acquire the quality of the water at water reservoirs, water beds, and at water tanks. Therefore, with the help of the proposed methodology, the concerned authorities will get the alerts at any time and from any place automatically if any unwanted levels are measured at the water sources. The complete system flow is clearly represented in shown in figure 2. Once the sensors share the data with the edge-level controlling device, the device will verify the values and compare these values with the standard values. Accordingly, the respective condition decision has been taken by the system.

III. LITERATURE SURVEY

A literature survey of IoT-based water contamination monitoring systems reveals a growing interest in this technology among researchers and practitioners. Many studies have focused on the development and testing of different types of sensors and communication systems for water quality monitoring, as well as the integration of these systems into a comprehensive water quality monitoring network. Maintaining and monitoring the quality of the water is essential nowadays for better livelihood. Because of this lot of research is going on in this domain and many researchers throughout the globe are working on this issue and are trying to develop a novel and cost-effective product, which is reachable to common man. With this inspiration we trying to develop a better product and during our literature survey we found similar works related to water quality monitoring are available in the literature, but they have their individual limitations. To overcome these limitations, we proposed this system. R. Vijayalakshmi [9] proposed a wonderful method in which lot of complex electronic circuits are involved and it is used for measuring the contamination in water based on retrieved sensor values. Jyotirmaya Ijaradar et.al [10] designed the water parameters measurement device with the help of Raspberry Pi, which is an efficient device, but relatively costly controller. K. Spandana et.al [11] developed a product that is designed with ESP 8266 WIFI module and system, not mention clearly which cloud server is being used. Other researchers [18,19,20,21] contributed similar works in which they implemented systems with IoT and WSN technologies.

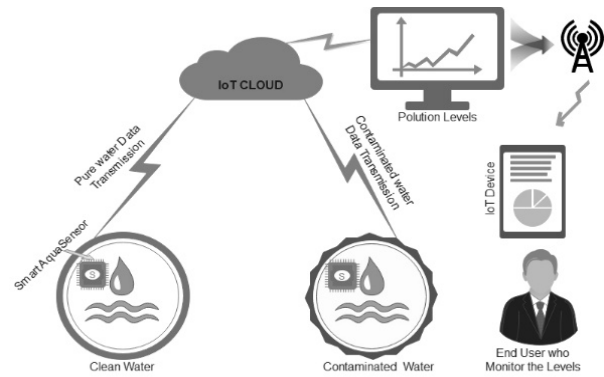


Fig. 1. Smart Water pollution Monitoring Systems with Smart Aqua sensors and IoT devices.

IV. METHODOLOGY

The main objectives of this work are: To build a real-time automated system that generates an alert whenever the water gets contaminated. To build a system that is feasible to use completely by the operators by showing the various parameters constantly anywhere and anytime. The architecture of the proposed system is shown in figure 3 in which the turbidity, pH value, and temperature values of the water sample are measured with respective sensors. The sensed values are fed to the microcontroller, which in turn compares the values with the respective threshold ranges. Based on the checked conditions mentioned in the systems flow diagram (Figure 2), the water quality status is sent to the concerned authorities via the ThingSpeak cloud interface [12,13,16]. The experimental set of the entire hardware and the measurement of murky turbidity water is depicted in figures 4 and 5,

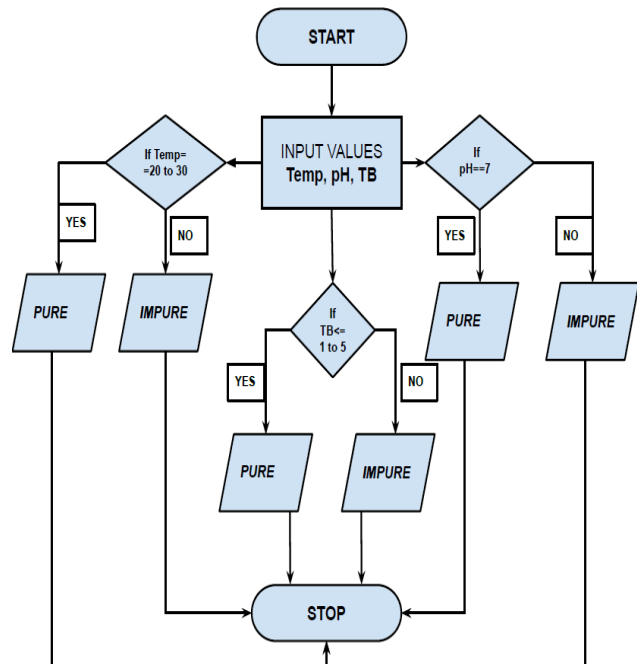


Fig. 2. System flow diagram

The proposed system's block diagram is shown in figure 3.

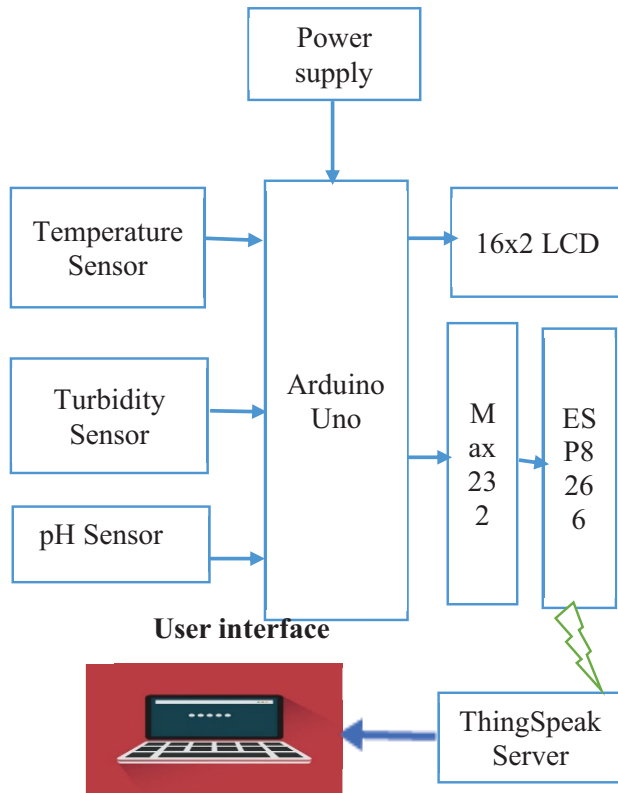


Fig. 3. System block diagram

EXPERIMENTAL SETUP

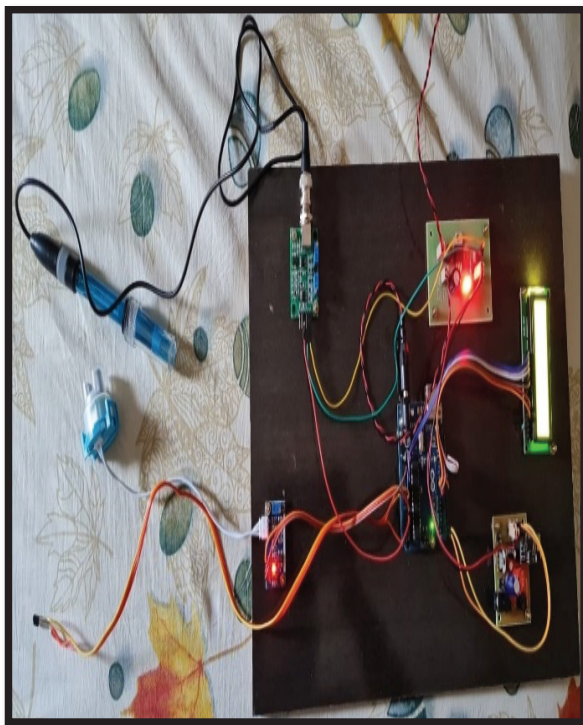


Fig. 4. Experimental Setup

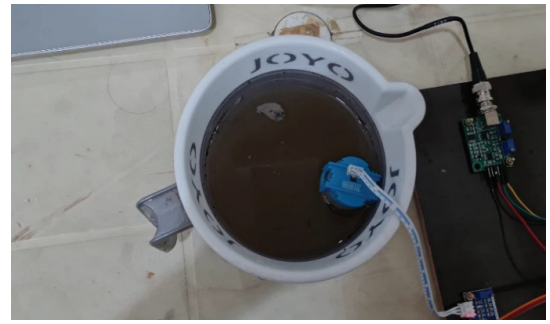


Fig. 5. Measuring pH of Murky water test in the experimental setup

V. RESULTS AND DISCUSSIONS

Temperature, pH, and turbidity are three parameters that are often used to determine the purity of a water sample. Temperature can be used as an indication of the overall health of a water body. For instance, extreme temperatures in water can cause stress to aquatic organisms and may indicate that the water is not pure. High or low temperatures can also be an indicator of water quality. Since the pH is a measure of the acidity or alkalinity of a water sample, a pH level that is outside of the normal range for a particular water body can indicate the presence of pollutants or other contaminants. Turbidity is a measure of the clarity of the water. High turbidity can be caused by the presence of particles or microorganisms in the water, which can indicate that the water is not pure. With the help of the proposed system, the above said three parameters are determined at different conditions and the respective results are depicted in Table 1. By analyzing the temperature, pH, and turbidity of a water sample, it is possible to determine whether the water is pure or impure. If the values for these parameters are within the normal range for a particular water body, and there are no other obvious signs of contamination, the water is likely to be pure. Since the pH, temperature, and turbidity values found by this system lie in the specified ranges as per the standards, the performance of the system is satisfactory. However, if any of these values are outside of the normal range or there are other indications of contamination, the water is likely to be impure.

Table 1. Results found on different parameters tested

Test ID	Item to be tested	Test Parameter	Expected Outcome	Actual Outcome
1	Warm water	Temperature in degree Celsius	45-55	51
2	Water at room temperature	Temperature in degree Celsius	25-30	32
3	Ice is added to the water	Temperature in degree Celsius	10-15	15
4	Distilled water	pH level in a pH scale	6.5-7.5	7
5	Lime juice is added	pH level in a pH scale	1-6	3
6	Soap water is added	pH level in a pH scale	8-14	10
7	Potable water	Turbidity Value (in NTU -nephelometric turbidity unit)	1-5	2
8	Murky (unpotable)	Turbidity Value (in NTU -nephelometric turbidity unit)	>5	8

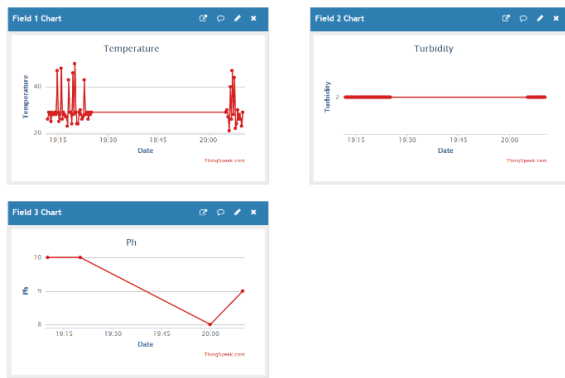


Fig. 6. Results in ThingSpeak Cloud Server.

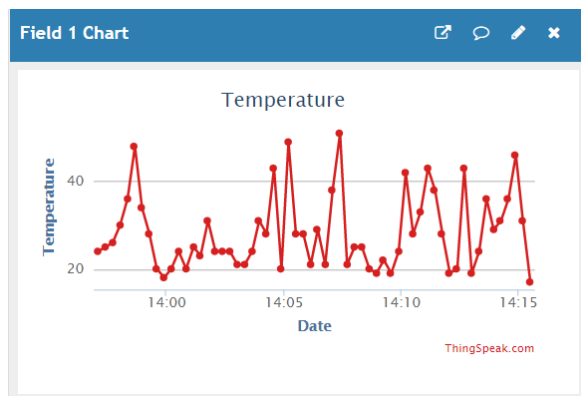


Fig. 7. Temperature Variations

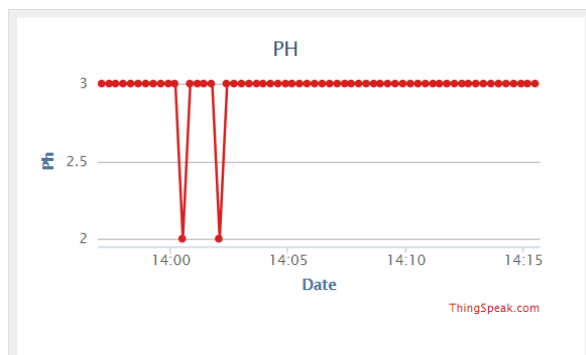


Fig. 8. Measured pH Value

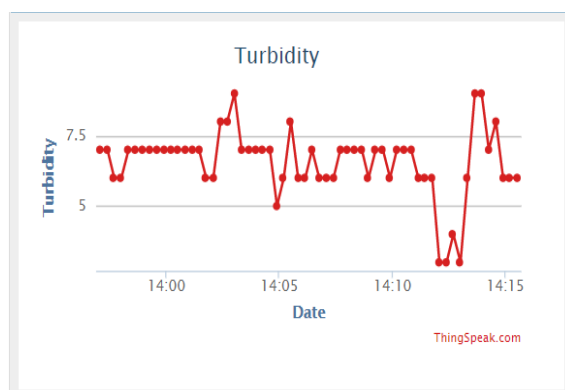


Fig. 9. Measured Turbidity values

Figure 6 describes how the turbidity, and pH of the water, are changing at room temperature. All the data are captured at the same time and sent to the cloud server. Also, the three sensors' data populated in the Thingspeak Cloud server are represented in a graphical manner and are separately shown in Figures 7,8 and 9.

VI. CONCLUSIONS AND FUTURE WORK

Water contamination is a major issue. This system helps to detect contamination levels in the water. This helps in saving time, money, and manpower for the organizations which are meant for maintaining the water body units like municipalities. The operator of the said organization can monitor the process online through the Thingspeak server and via an interface setup. This setup has been implemented with the IoT and embedded system principles, and it is definitely a potential solution to the existing manual system and will tremendously reduce the difficulties faced during the measurement of water contamination levels at different water beds. Since this proposed system is one of the important modules of any Smart City application, this can be further enhanced by including more sensors related to the chemical and biological characteristics of the water.

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